A. Travail sous Matlab

```
% Travail demandé 1:

    clear all;
    close all;
    clc;



    t = 0:0.01:50;
    f=sin(t);
    g=sin(t/2);
    plot(t,f, '');
    hold on;
    plot(t,g, 'g');
    hold off;
    xlabel('t');
    ylabel('y');
    title('fonction sin(t) et sin(t/2)');
    legend('sin(t)','sin(t/2)');

% Travail demandé 2:

    H1=tf([1 1],[1 2]);
    H2=tf([1 2 3],[1 4 5 6]);
    H3=tf([3 0 0],[2 6 3]);

% Travail demandé 3:

    G=zpk([0],[2 (1+i) (1-i)],-2);

    w=zpk([-1 -3],[0 -2 -4],1);

% Travail demandé 4:

    g=tf(10,[1 5 3]);
    h=zpk([-4],[-1 -2], 1);

    sys=feedback(g, h);

    % methode 1

        g=tf(1, conv([1 5],[1 2]));

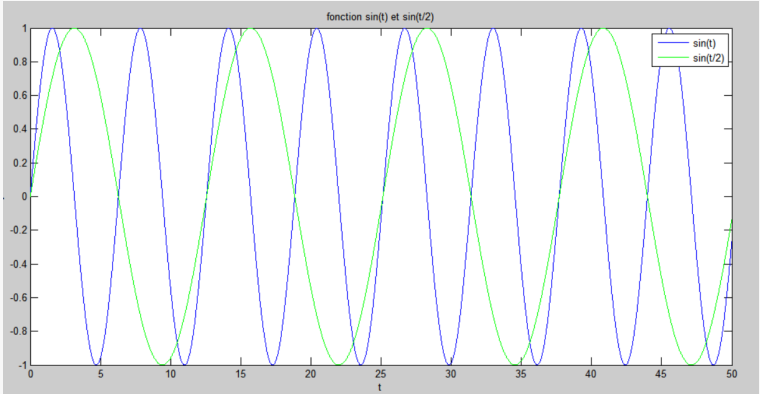
    % methode 2

        g=zpk([],[-5 -2], 1);

    sys=feedback(g, 1);

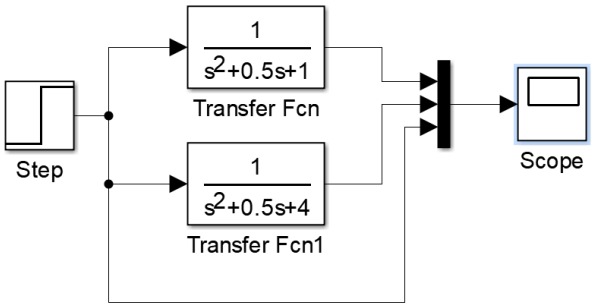
% Travail demandé 5:

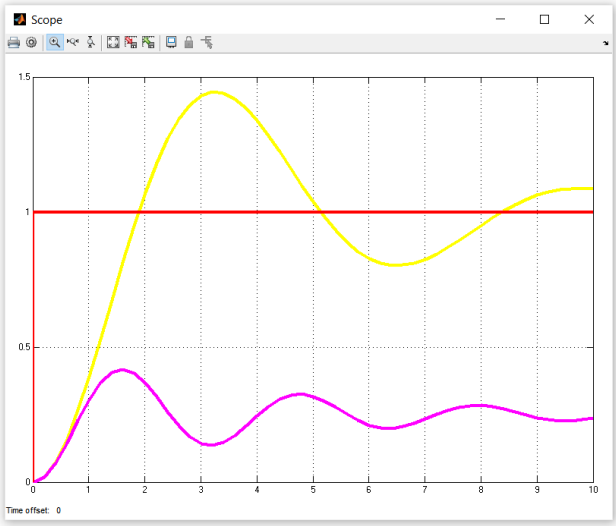
    num = [1 3];
    den = [1 3 2];
    [r, p, k] = residue(num,den);
```



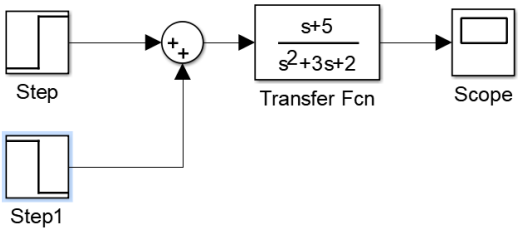
B) Travail sous SIMULINK

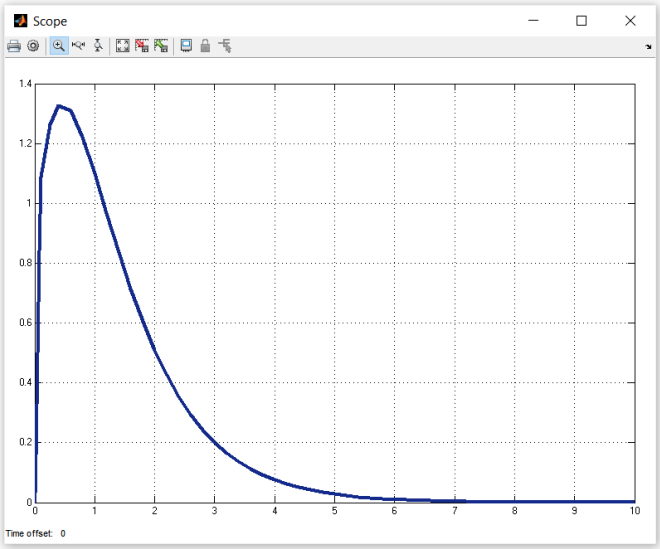
a) Réponse indicielle avec Simulink





b) Réponse impulsionnelle avec Simulink





c) Réponse à une rampe avec Simulink

